

Multi-Fidelity Verification & Physical Signoff Funnel

Structural & Functional Failures
Route to Generative Prompt / AST Repair

Generative Proposals (Candidate Pool N_0)

Stage 1: Syntactic & AST Parsing

Checks: Syntax valid, elaborated module hierarchy, port types
Engines: Tree-Sitter / Verilator / Slang • Cost: Milliseconds (free / open-source)

Stage 2: Interface Schema & Interconnect

Checks: AXI / TLM handshakes, port widths, clock domain tags
Engines: Static Schema Linters / Protocol Checkers • Cost: Seconds (low compute overhead)

Stage 3: Functional & Assertion Verification

Checks: Temporal invariants, state-machine deadlocks, coverage
Engines: SystemVerilog Assertions (SVA) / BMC / Sim • Cost: Minutes (simulation pool / SAT solver)

Stage 4: Static Timing Analysis (STA)

Checks: Multi-corner setup & hold slack, WNS/TNS, max transition
Engines: OpenSTA / Synopsys PrimeTime / Cadence Tempus • Cost: Tens of minutes (licensed STA seat)

Stage 5: Physical DRC / LVS Signoff

Checks: Design rule clean (DRC), layout vs. schematic (LVS)
Engines: OpenROAD / Cadence Innovus / Synopsys ICC2 • Cost: Hours to days (full signoff toolchain)

Tapeout-Qualified Clean Implementation (GDSII / OASIS)

Signoff Funnel Economics:

- Hierarchical Multi-Fidelity:
Fast AST & schema checks prune non-viable candidates before licensed tools run.
- License Capacity Protection:
Commercial STA & PnR tools are strictly rate-limited by license pool seats.
- Separation of Concerns:
Architectural search proposes candidate RTL; ECO closures repair physical margins without restarting the generative loop.

Near-Miss Timing & DRC Failures
Route to Localized ECO Optimization
(Buffer insertion, gate sizing, wire widening)